Arista Access Node KR-ACC-24 QoS Congestion Event Resolution SOP

Node: KR-ACC-24 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_5386

Description

The KR-ACC-24 node at KR Puram Access site is an Arista access node operating in QoS-enabled mode, employing the Multi-Instance Queuing (MIQ) and Priority Flow Control (PFC) mechanisms to manage packet flows. The node's interface, likely an SFP+ or QSFP28 module, may experience microbursts or interface tail drops due to the QoS queue starvation caused by the bursty traffic patterns. This phenomenon can lead to packet drops and subsequent congestion events, triggering the ALARM_5386 alarm code. As the Arista access node relies on the egress queue management to mitigate the effects of congestion, the node's performance degradation can be attributed to the inadequate QoS configuration, misaligned buffer sizes, or faulty hardware. Therefore, this SOP aims to diagnose and rectify the root cause of the QoS congestion event on the KR-ACC-24 node.

Pre-requisite Checks

- Check: Verify the site location and the affected node ID, KR-ACC-24.

- Check: Confirm the vendor and the CLI dialect in use for this node, Arista, and ensure the correct CLI commands are executed.

- Check: Establish remote access to the node and execute privileged mode commands.

- Check: Verify the redundant link status and ensure it is functioning properly.

Potential Root Causes

Cause	Probability
Misaligned buffer sizes or inadequate QoS configuration.	High
Faulty hardware or interface module.	Medium
Malfunctioning or misconfigured PFC or MIQ mechanisms.	Low

Diagnostic Commands & Expected Outputs

```
Command: show queue statistics, Expected Output: Queue statistics for egress queues, including packet drops and buffer utilization., Description: This command will provide insight into the packet drops and buffer utilization on the egress queues, helping to identify potential congestion points and QoS configuration issues.

Command: show interface statistics, Expected Output: Interface statistics, including error counts and packet drops., Description: This command will provide information on the interface's error counts and packet drops, which can aid in identifying hardware-related issues or misconfigured QoS settings.
```

```
Command: show qos configuration, Expected Output: QoS configuration, including class maps and policy maps., Description: This command will display the current QoS configuration, allowing for the identification of potential misaligned buffer sizes or inadequate QoS settings.
```

Resolution Steps

Step 1: Step: Adjust QoS Configuration, Command: config qos class-map match-cos 0, Description: Modify the QoS class map to adjust the buffer size and packet drop thresholds to mitigate congestion events.

Step 2: Step: Verify PFC Configuration, Command: show pfc configuration, Description: Verify the PFC configuration to ensure it is properly aligned with the QoS settings and buffer sizes.

Step 3: Step: Replace Faulty Hardware (if applicable), Command: no interface , Description: Remove the faulty interface module and replace it with a new one, ensuring proper configuration and alignment with the existing QoS settings.

Post-Check Verification

```
Command: show queue statistics, Expected Output: Queue statistics indicating reduced packet drops and buffer utilization., Description: Verify that the QoS configuration adjustments and PFC verification have resolved the congestion event by checking the egress queue statistics.

Command: show interface statistics, Expected Output: Interface statistics indicating reduced error counts and packet drops., Description: Verify that the resolution steps have resolved the congestion event by checking the interface statistics.
```

Roll-back Procedures

- Step: Restore Original QoS Configuration, Command: config qos class-map match-cos 1, Description: Restore the original QoS configuration to its previous state, reverting any changes made during the resolution process.

- Step: Re-configure PFC (if applicable), Command: config pfc , Description: Re-configure the PFC settings to their original state, ensuring proper alignment with the QoS settings and buffer sizes.